

Measuring the Size of Charon

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How can we measure the size of Pluto's moon, Charon, when it subtends only one twentieth of an arcsec? Imaging, even with adaptive optics, cannot resolve Charon. One solution was to observe Pluto and Charon during Pluto's equinox, when the two bodies eclipsed and transited each other, but this method gave conflicting measurements ranging from 593 to 620 km. A more direct method is stellar occultation, when Charon passes in front of a star as seen from observers within the shadow path on Earth. Knowing Charon's velocity, a duration for the star's disappearance can be turned into a chord length. Combining observations from multiple sites gives a raster scan across Charon, revealing its shape and size. One Charon occultation had been observed previously, by Alistair Walker in 1980, but with one chord across Charon this only set a lower limit to the radius.

An accurate value for Charon's radius would improve our understanding of the Pluto-Charon system. Charon's energy budget and spectral modeling depend on its radius through its albedo. Charon's surface gravity, escape velocity, and density depend on its radius, which affects our models of Charon's composition and formation; the current uncertainty in Charon's radius allows for Charon densities ranging from 1.6 to 1.8 g/cm³. Finally, an occultation is capable of detecting even a tenuous atmosphere, if one is present.

For the event which lasted only less than a minute, they chose a Princeton Instruments 512x512 back illuminated, frame transfer CCD camera as it has no dead time between images. The camera was mounted with no re-imaging optics, so our 13-micron pixels gave a field of view of only 21 arcsec. For the event, the camera was operated with 0.2-sec integrations from the GPS-slaved timing unit, binning 4x4 on-chip for an effective plate scale of 0.16"/pixel.

Within half an hour of the event, the team reported a successful observation with results confirmed by data from two other instruments. Results shown below.

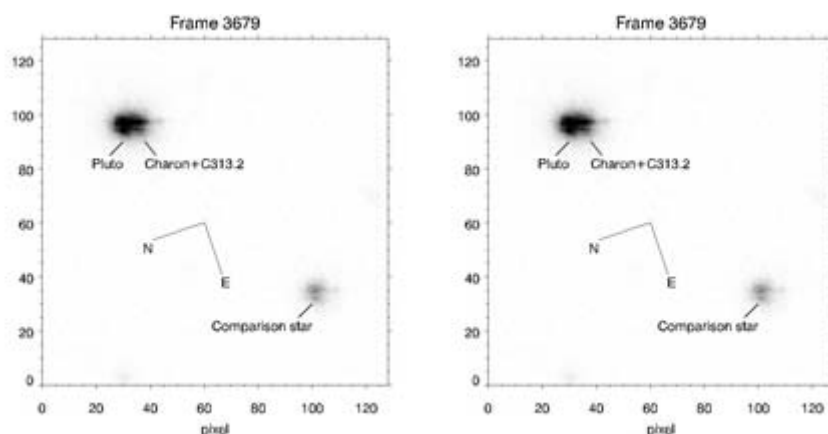


Figure 1. Two frames from the event sequence showing Pluto-Charon-C313.2 and a comparison star.



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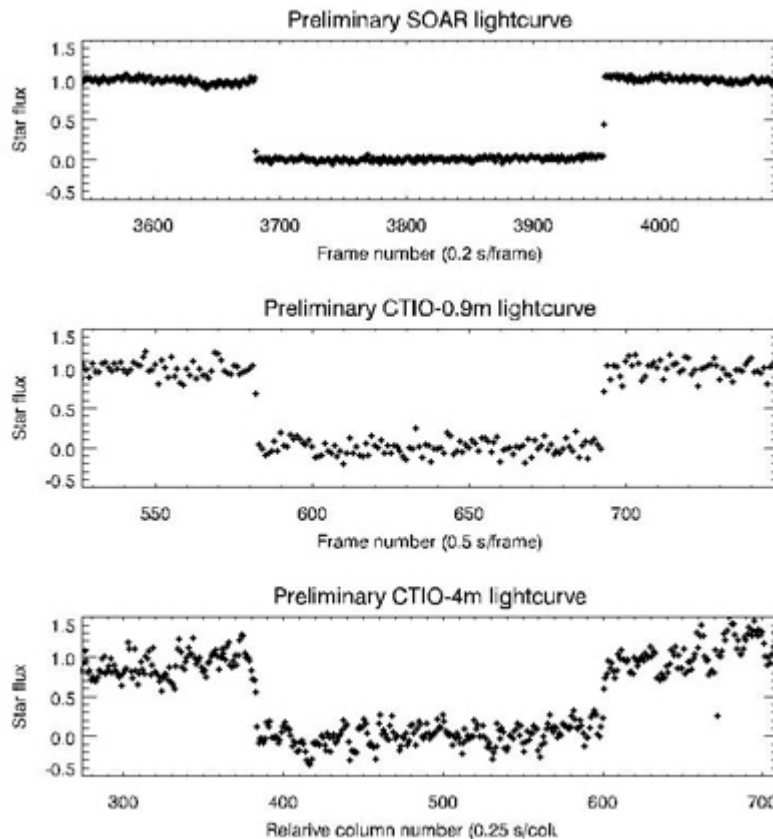


Figure 2. Light curves from three different observations of Charon event show that the event lasted for about 55 seconds and no evidence of atmosphere. Data courtesy of Eliott Young and Leslie Young, Southwest Research Institute, Boulder Colorado, USA.

The research team at SWRI recently acquired multiple PhotonMAX cameras for observation of Pluto's occultation during March 18, 2007. The cameras offer all the advantages of frame transfer cameras for these time resolved photometry application in a compact (no controller) package.

Frame transfer CCDs for time resolved astronomy

While typical astronomical observations use minutes to hours of integration times, planetary and stellar motion studies require only sub-second exposures. One of the requirements for such observations is 100% duty cycle imaging i.e., no dead time between exposures to get the highest time resolution. Princeton Instruments offer several different frame transfer cameras for the applications.

PhotonMAX: 512B and 1024B EMCCD cameras offer several advantages:

- Both EM and traditional CCD operation modes
 - Back illuminated, frame transfer architecture for high sensitive 100% duty cycle imaging.
 - High frame rate from 10 fps - 30 fps at full resolution
 - 1MHz digitization (traditional) for low read noise. Useful, when the signal is above the read noise as it does not introduce additional noise factor as in EM mode.
- o Deep cooling down to -90C
 - o Liquid cooling with no-fan option to eliminate vibration and thermal air currents around telescopes
 - o Compact size and controller-less operation
 - o Fiber optic data interface for remote operation
 - o Circular buffers for capturing long sequences

Note: A traditional back illuminated, frame transfer 1024x1024 CCD is available in VersArray (and PIXIS) platforms.



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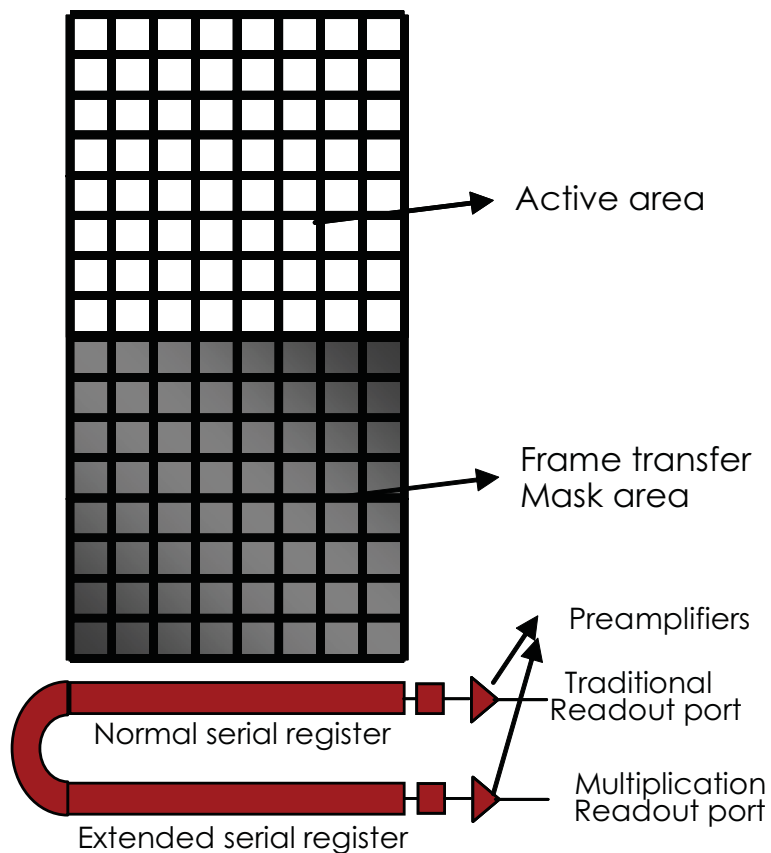


Figure 3. PhotonMAX: 512B and 1024B frame transfer EMCCD cameras offer both traditional and EM (multiplication) gain ports.



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